Table 10.1 Ionic Channels in the Axolemma

Leakage Channels are ionic channels that remain open all the time	K^+ leakage channels are more in number than the Na^+ leakage channels. Sarcolemma has greater permeability to K^+ ions than Na^+ ions. These ions keep moving continuosly maintain the potential difference across the axolemma.
Ligand-gated Channels are chemically gated channels which open or close in response to a chemical stimuli.	They are located between the presynaptic membrane of the first axon and post synaptic membrane of the cell body of second neuron [i.e. dendrites and cell bodies]. The neurotransmitter acetylcholine opens ligand channels that allow Na^+ and Ca^{++} ions diffuse inward and K^+ ions diffuse outward.
Voltage-gated Channels are mechanically gated channels which open in response to a physical stimulus in the form of vibration such as touch and pressure.	These channels open in response to a change in membrane potential. There are two types of voltage-gated channels. **i. Sodium voltage-gated channel** **ii. Potassium voltage-gated channel**

Na^+ influx into the cell. When the axon is not conducting any impulses i.e. in resting condition, the axon membrane is more permeable to K^+ and less permeable to Na^+ ions, whereas it remains impermeable to negatively charged protein ions.

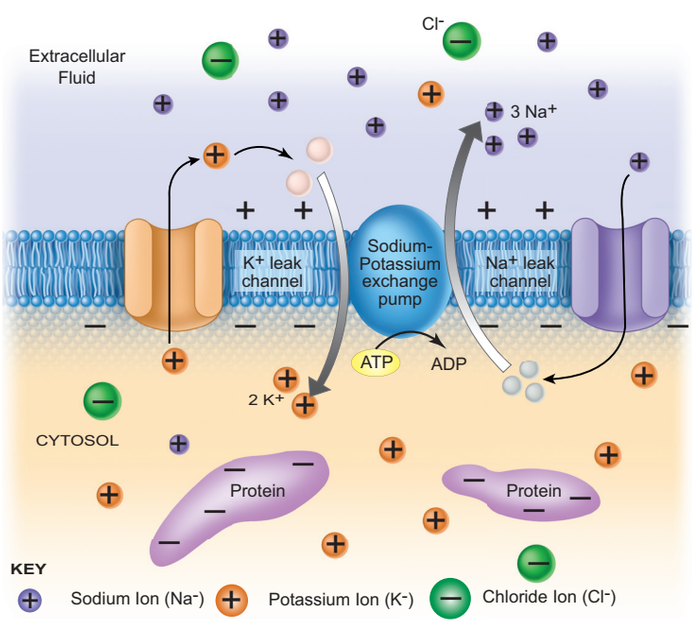


Figure 10.3 Ionic channels

The axoplasm contains high concentration of K^+ and negatively charged proteins and low concentration of Na^+ ions. In contrast, fluid outside the axon (ECF) contains low concentration of K^+ and high concentration of Na^+, and this forms a concentration gradient. This ionic gradient across the resting membrane is maintained by ATP driven **Sodium-Potassium** pump, which exchanges $3Na^+$outwards for $2K^+$ into the cells. In this state, the cell membrane is said to be **polarized**. In neuron, the **resting membrane potential** ranges from **-40mV** to **-90mV**, and its normal value is **-70mV**. The minus sign indicates that the inside of the cell is negative with respect to the outside (Figure 10.4).

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